Scholten cover algebra and exceptional fibres 2026-10-31

T-1127. Producer packet; independent source-fidelity review pending.

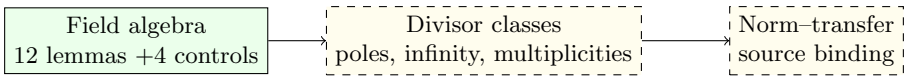
Checked component

Twelve field-algebra declarations and four fixed control propositions compiled in Lean. Separate readback of all16 declarations reports only standard axioms (propositional extensionality, quotient soundness, and classical choice where used). Source/input bindings remained unchanged. No independent binary/kernel rechecker was run.

The implemented forms are $A = (1+i)t + i$, $B = (1-i)t - i$, $i^2 = -1$. Checked identities include the determinant $-2i$, the quadratic leading coefficient $2i(1+x)$, conjugation exchanging A, B, denominator normalization, and fibre signs. In particular,

$$A(t)^2 + B(t)^2 = -4t - 2.$$

Thus the $x = -1$ fibre requires an infinity contribution when represented on the quintic. The checks require their stated nonzero denominators; they do not certify the Rust field implementation.



Geometric route, conditional on class bindings

Put $z = A/B$, $Y = y/B^3$. The two elliptic quotients are

$$f(z, Y) = (z^2, Y), \qquad g(z, Y) = (z^{-2}, Y/z^3).$$

The g-fibre involution $(z, Y) \mapsto (-z, -Y)$ negates the f-image. On smooth proper models, the resulting cross push–pull maps vanish, while the diagonal degree-two compositions equal [2]. This gives the usual product-to-Jacobian isogeny; conjugation exchanges its factors, supplying the proposed descent from the Weil restriction.

After binding the literal divisor representatives, $f_*\tau(P) = [2]P$ implies nonzero transfer on a nonzero odd-prime-order point. A base-field isogeny then identifies the Jacobian and Weil-restriction point counts. These are written geometric deductions with explicit obligations, not conclusions certified by the twelve algebra lemmas alone.

Exceptional cases to bind

At the quintic point at infinity, $f(\infty) = T = (-1, 0)$. A degree-d Mumford class pushes to $\sum f(Q_j) - dT$. The source subtracts T for degree one; degree two needs no correction because $2T = O$. For $x = -1$, the finite point is $W = (-1/2, 0)$ and its representative is $W - \infty$. At $B = 0$, the source returns elliptic infinity. At $A = 0$, ramification can give repeated roots whose multiplicities must be preserved. These source/class identifications, Cantor reduction and projective extension remain to be certified.

Evidence and sources

The first proof attempt failed on a redundant tactic; attempt2 passed. Four controls and separate axiom readback passed, with all attempts retained. Source: `three_leads.rs`, lines1094–1247. No scientific execution, timing, or cost ratio.

Diem, genus-two covers of degree two, Section2 and Proposition2.14, supplies classical complementary-cover context (2 invertible). Milne, Abelian Varieties, I10.20 and II1.1, supplies the characteristic-polynomial/point-count implication once an isogeny is bound. No novelty claim. Full derivation, literal obligations, hashes and statuses accompany this PDF.